Carter Perez

Annapolis, MD | carterperez44@gmail.com | (410) 212-7288 | GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Software engineer with extensive full-stack experience and backend specialization, currently developing competition infrastructure for CISA's President's Cup CTF. Core expertise in API design, authentication systems, distributed architecture, and containerized production deployments.

TECHNICAL SKILLS

Languages: Python, Go, Rust, TypeScript, JavaScript, Haskell
Backend: FastAPI, Flask, Axum (Rust), Chi/Gin (Go), Node.js
Frontend: React, SolidJS, TypeScript, SCSS, Tailwind
Databases: PostgreSQL, MongoDB, Redis, SurrealDB
Infrastructure: Docker, Nginx, GitHub Actions, AWS, Linux

PROJECTS

CertGames | certgames.com | GitHub

- Developed full-stack cybersecurity learning platform and iOS app with 1,000+ users, gamification engine with XP/achievements, leaderboards, and 15K+ questions across 14 certifications
- Developed enterprise SaaS combining React/TypeScript with Flask backend across 20+ DDD modules, decorator composition for middleware, Pydantic validation, async MongoDB/Redis operations, automated CI/CD, and self-hosted deployment with Cloudflare Tunnels

Encrypted P2P Chat | GitHub

- Architected end-to-end encrypted messaging platform implementing Signal Protocol with X3DH key exchange and Double Ratchet algorithm, leveraging triple-database architecture with WebAuthn passwordless authentication and SolidJS frontend

docksec - Docker Security Scanner | GitHub

- Built enterprise CLI security scanner in Go validating Docker environments against 100+ CIS Docker Benchmark controls with multi-format reporting, checking privileged containers, dangerous capabilities, sensitive path mounting, and security profile compliance

Production FastAPI + React Template | GitHub

- Built production-ready boilerplate with JWT authentication featuring automatic token rotation, SafeEnum pattern preventing database breakage on refactors, timing-safe password verification preventing user enumeration, and token family tracking detecting replay attacks

PROFESSIONAL EXPERIENCE

Software Engineer

Cyber Skyline | *Remote* | *May 2025 – Present*

- Backend engineer for CISA President's Cup CTF developing core competition systems including scoring engine, notifications, support ticketing, and team/event management - #1 contributor
- Engineered notification system coordinating multi-instance Flask deployments via Redis pub/sub, implementing per-instance user session tracking to route messages across horizontally scaled containers without shared state
- Developed composable decorator middleware for authentication, resource loading, and error handling - standardizing security patterns and eliminating boilerplate across 60+ endpoints

Integration Tech II

Sealing Technologies | *Columbia, MD* | *May 2024 – May 2025*

- Engineered quality assurance automation for $40M government server deployment, developing Python scripts to validate hardware configurations and reduce manual testing cycles by 30%
- Architected standardized assembly SOPs adopted across 5+ teams, documenting technical specifications for custom server builds meeting strict regulatory compliance requirements

CERTIFICATIONS

CompTIA SecurityX, PenTest+, CySA+, Security+, Network+, A+ | Python Institute PCEP

EDUCATION

University of Maryland Global Campus | Expected May 2027
M.S. Cybersecurity (In Progress) | Largo, MD